

Prayas JEE 3.0 2025

Maths

DPP: 4

Trigonometric Functions

Q1 $(1 + \cos \frac{\pi}{8})(1 + \cos \frac{3\pi}{8})(1 + \cos \frac{5\pi}{8})(1 + \cos \frac{7\pi}{8})$ is equal to

- (A) $\frac{1}{2}$
 (B) $\cos \frac{\pi}{8}$
 (C) $\frac{1}{8}$
 (D) $\frac{1+\sqrt{2}}{2\sqrt{2}}$

Q2 $\cos \frac{\pi}{11} + \cos \frac{3\pi}{11} + \cos \frac{5\pi}{11} + \cos \frac{7\pi}{11} + \cos \frac{9\pi}{11}$ is equal to

- (A) 1
 (B) $\frac{1}{2}$
 (C) $\frac{1}{4}$
 (D) 2

Q3 If $1 + \sin \theta + \sin^2 \theta + \dots \infty = 4 + 2\sqrt{3}$, $0 < \theta < \pi$, $\theta \neq \frac{\pi}{2}$, then

- (A) $\theta = \frac{\pi}{3}$
 (B) $\theta = \frac{\pi}{6}$
 (C) $\theta = \frac{\pi}{3}$ or $\frac{\pi}{6}$
 (D) $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

Q4 Given that $\tan A$ and $\tan B$ are the roots of the equation $x^2 - ax + b = 0$. The value of $\sin^2(A + B)$, is

- (A) $\frac{a^2}{a^2 + (1-b)^2}$
 (B) $\frac{a^2}{a^2 + b^2}$
 (C) $\frac{a^2}{(a+b)^2}$
 (D) $\frac{a^2}{b^2 + (1-a)^2}$

Q5 The expression

$$2 \sin 2^\circ + 4 \sin 4^\circ + 6 \sin 6^\circ + \dots$$

$180 \sin 180^\circ$ equals

- (A) $\cot 1^\circ$
 (B) $90 \cot 1^\circ$
 (C) $\sin 1^\circ$
 (D) $90 \cos 1^\circ$

Q6 $\sum_{r=1}^n \left(\frac{1}{\cos \theta + \cos(2r+1)\theta} \right)$, $n \in \mathbb{N}$ is equal to

- (A) $\frac{\sin(n+1)\theta}{\sin \theta \cdot \cos n\theta}$
 (B) $\frac{\sin n\theta}{\sin 2\theta \cdot \cos(n+1)\theta}$
 (C) $\frac{\tan(n+1)\theta}{\sin n\theta}$
 (D) $\frac{\sin(n-1)\theta}{\sin \theta \cdot \cos n\theta}$

Q7 If $4n\alpha = \pi$, then

$\cot \alpha \cdot \cot 2\alpha \cdot \cot 3\alpha \dots \cot(2n-1)\alpha$ is equal to

- (A) 1
 (B) -1
 (C) 2
 (D) Zero

Q8 Let $f(x) = \sin^4 x + \cos^4 x$. Then f is an increasing function in the interval

- (A) $[0, \frac{\pi}{4}]$
 (B) $[\frac{\pi}{4}, \frac{\pi}{2}]$
 (C) $[\frac{\pi}{2}, \frac{5\pi}{8}]$
 (D) $[\frac{5\pi}{8}, \frac{3\pi}{4}]$

Q9 If $\alpha + \beta + \gamma = \pi$, the

$$\sin^2 \alpha + \sin^2 \beta - \sin^2 \gamma =$$

- (A) $2 \sin \alpha \sin \beta \cos \gamma$
 (B) $2 \cos \alpha \cos \beta \cos \gamma$
 (C) $2 \sin \alpha \sin \beta \sin \gamma$
 (D) None of these

Q10



If $\cos(\theta - \alpha)$, $\cos \theta$ and $\cos(\theta + \alpha)$ are in H.P.,
then

- (A) $\sin \theta = \sqrt{2} \cos\left(\frac{\alpha}{2}\right)$
- (B) $\cos \theta = \sqrt{2} \cos\left(\frac{\alpha}{2}\right)$
- (C) $\cos \theta = \sqrt{2} \sin\left(\frac{\alpha}{2}\right)$
- (D) $\sin \theta = \sqrt{2} \sin\left(\frac{\alpha}{2}\right)$



Answer Key

Q1 (C)

Q2 (B)

Q3 (D)

Q4 (A)

Q5 (B)

Q6 (B)

Q7 (A)

Q8 (B)

Q9 (A)

Q10 (B)



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